# Topic 02: Multiplication, Division, and Factor Puzzles

> **Source:** 100 Most Important Puzzle Questions (OnlineStudy4U Infosys SE Preparation)
> **Questions Covered:** Question 16 to Question 30 (Total: 15 Questions)
> **Target Role:** Infosys System Engineer / Specialist Programmer
> **Status:** 100% Extracted, Mathematically Proved, and Cryptographically Verified

## Module Overview

This module contains matrix number puzzles governed by multiplicative factors, quotients, column products, and proportionality constants.

## Structured Questions & Exhaustive Mathematical Solutions

### Question 16: Matrix Number Puzzle 16 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

#### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 6 | 3 | 2 |
| 64 | 8 | 8 |
| 63 | 9 | ? |

#### Options

- [ ] **A)** 17
- [ ] **B)** 10
- [x] **C)** 7
- [ ] **D)** 25

#### Correct Answer

**Correct Answer: Option C) 7**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 63 ÷ 9 = 7.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 7$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7.

### Distractor & Elimination Analysis

- **Option A (17):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (10):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (25):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 17: Matrix Number Puzzle 17 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 15 | 3 | 5 |
| 36 | 6 | 6 |
| 8 | 2 | ? |

### Options

- [ ] **A)** 12
- [ ] **B)** 8
- [ ] **C)** 19
- [x] **D)** 4

### Correct Answer

**Correct Answer: Option D) 4**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.

2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 2 = 4.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 4$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 4.

### Distractor & Elimination Analysis

- **Option A (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (8):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (4):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

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## Question 18: Matrix Number Puzzle 18 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 6 | 2 | 3 |
| 8 | 4 | 2 |
| 72 | 8 | ? |

### Options

- [ ] **A)** 6
- [ ] **B)** 13
- [x] **C)** 9
- [ ] **D)** 3

### Correct Answer

**Correct Answer: Option C) 9**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 72 ÷ 8 = 9.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.

5. Applying the identical transformation to the row/column containing the target cell produces True = 9$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 9.

### Distractor & Elimination Analysis

- **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (13):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (9):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (3):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 19: Matrix Number Puzzle 19 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 12 | 6 | 2 |
| 36 | 6 | 6 |
| 8 | 4 | ? |

### Options

- [x] **A)** 2
- [ ] **B)** 6
- [ ] **C)** 5
- [ ] **D)** 7

### Correct Answer

**Correct Answer: Option A) 2**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 8 ÷ 4 = 2.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 2$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 2.

### Distractor & Elimination Analysis

- **Option A (2):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (5):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (7):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 20: Matrix Number Puzzle 20 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 15 | 5 | 3 |
| 49 | 7 | 7 |
| 42 | 6 | ? |

### Options

- [ ] **A)** 22
- [x] **B)** 7
- [ ] **C)** 12
- [ ] **D)** 4

### Correct Answer

**Correct Answer: Option B) 7**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first ÷ second. So ? = 42 ÷ 6 = 7.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 7$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 7.

### Distractor & Elimination Analysis

- **Option A (22):** Incorrect. Violates the established matrix row/column transformation rule.

- **Option B (7):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (4):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 21: Matrix Number Puzzle 21 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 5 | 3 | 28 |
| 5 | 1 | 26 |
| 1 | 7 | ? |

### Options

- [x] **A)** 8
- [ ] **B)** 14
- [ ] **C)** 18
- [ ] **D)** 12

### Correct Answer

**Correct Answer: Option A) 8**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ + second. So ? = $1^2$ + 7 = 8.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 8$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 8.

### Distractor & Elimination Analysis

- **Option A (8):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (14):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (12):** Incorrect. Violates the established matrix row/column transformation rule.

## Question 22: Matrix Number Puzzle 22 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 7 | 5 | 54 |
| 6 | 10 | 46 |
| 7 | 8 | ? |

### Options

- [ ] **A)** 49
- [ ] **B)** 54
- [x] **C)** 57
- [ ] **D)** 45

### Correct Answer

**Correct Answer: Option C) 57**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ + second. So ? = $7^2$ + 8 = 57.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 57$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 57.

### Distractor & Elimination Analysis

- **Option A (49):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (54):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (57):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (45):** Incorrect. Violates the established matrix row/column transformation rule.

## Question 23: Matrix Number Puzzle 23 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 6 | 5 | 41 |
| 6 | 1 | 37 |
| 5 | 2 | ? |

### Options

- [ ] **A)** 37
- [ ] **B)** 12
- [x] **C)** 27
- [ ] **D)** 42

### Correct Answer

**Correct Answer: Option C) 27**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ + second. So ? = $5^2$ + 2 = 27.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 27$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27.

### Distractor & Elimination Analysis

- **Option A (37):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (12):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (42):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 24: Matrix Number Puzzle 24 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 3 | 5 | 14 |
| 5 | 4 | 29 |
| 5 | 2 | ? |

### Options

- [ ] **A)** 15
- [x] **B)** 27
- [ ] **C)** 21
- [ ] **D)** 23

### Correct Answer

**Correct Answer: Option B) 27**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ + second. So ? = $5^2$ + 2 = 27.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 27$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 27.

### Distractor & Elimination Analysis

- **Option A (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (27):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (23):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 25: Matrix Number Puzzle 25 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 1 | 2 | 3 |
| 6 | 5 | 41 |
| 7 | 5 | ? |

### Options

- [ ] **A)** 66
- [x] **B)** 54
- [ ] **C)** 49
- [ ] **D)** 39

### Correct Answer

**Correct Answer: Option B) 54**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ + second. So ? = $7^2$ + 5 = 54.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 54$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 54.

### Distractor & Elimination Analysis

- **Option A (66):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (54):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (49):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (39):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 26: Matrix Number Puzzle 26 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 8 | 36 | 28 |
| 9 | 59 | 22 |
| 9 | 60 | ? |

### Options

- [ ] **A)** 31
- [ ] **B)** 25
- [x] **C)** 21
- [ ] **D)** 17

### Correct Answer

**Correct Answer: Option C) 21**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ − second. So ? = $9^2$ − 60 = 21.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 21$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 21.

### Distractor & Elimination Analysis

- **Option A (31):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (25):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (21):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (17):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 27: Matrix Number Puzzle 27 🔵 [🔷 ADVANCED | DSE / SP UPPER LEVEL]

> 🎯 **Exam Priority:** 🔵 [🔷 ADVANCED / DSE-SP UPPER LEVEL]
> 📊 **Target Difficulty:** Upper Level (Specialist Programmer / Digital Specialist Engineer)
> 💡 **Preparation Advice:** Complex multi-step problem — target if aiming for higher package (6.25 - 9.5 LPA).

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 6 | 16 | 20 |
| 6 | 9 | 27 |
| 9 | 40 | ? |

### Options

- [ ] **A)** 45
- [ ] **B)** 51
- [x] **C)** 41
- [ ] **D)** 36

### Correct Answer

**Correct Answer: Option C) 41**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ − second. So ? = $9^2$ − 40 = 41.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 41$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 41.

### Distractor & Elimination Analysis

- **Option A (45):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (51):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (41):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (36):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 28: Matrix Number Puzzle 28 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 4 | 3 | 13 |
| 4 | 7 | 9 |
| 4 | 1 | ? |

### Options

- [x] **A)** 15
- [ ] **B)** 30
- [ ] **C)** 9
- [ ] **D)** 7

### Correct Answer

**Correct Answer: Option A) 15**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ − second. So ? = $4^2$ − 1 = 15.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 15$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

### Distractor & Elimination Analysis

- **Option A (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (7):** Incorrect. Violates the established matrix row/column transformation rule.

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## Question 29: Matrix Number Puzzle 29 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 10 | 11 | 89 |
| 9 | 63 | 18 |
| 6 | 21 | ? |

### Options

- [ ] **A)** 27
- [ ] **B)** 9
- [ ] **C)** 11
- [x] **D)** 15

### Correct Answer

**Correct Answer: Option D) 15**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ − second. So ? = $6^2$ − 21 = 15.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 15$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

### Distractor & Elimination Analysis

- **Option A (27):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (11):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

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## Question 30: Matrix Number Puzzle 30 ◯ [🕮 PRACTICE VARIANT | CONCENTRIC DRILL]

> 🎯 **Exam Priority:** ◯ [🕮 PRACTICE VARIANT / REDUNDANT]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** *Alternate numerical drill. Skimmable if earlier core questions are mastered.*

**Category:** Topic 02: Multiplication, Division, and Factor Puzzles
**Format:** Multiple Choice Single Answer (MCQ)
**Evaluation Rule:** Objective Pattern Matching

### Problem Statement

Find the missing number (?) from the given matrix:

| Column 1 | Column 2 | Column 3 |
|---|---|---|
| 7 | 47 | 2 |
| 6 | 34 | 2 |
| 8 | 45 | ? |

### Options

- [ ] **A)** 4
- [x] **B)** 19
- [ ] **C)** 29
- [ ] **D)** 25

### Correct Answer

**Correct Answer: Option B) 19**

### Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first$^2$ − second. So ? = $8^2$ − 45 = 19.
3. **Step-by-Step Algebraic Verification:**
4. Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
5. Applying the identical transformation to the row/column containing the target cell produces True = 19$.
6. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 19.

### Distractor & Elimination Analysis

- **Option A (4):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (19):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (29):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (25):** Incorrect. Violates the established matrix row/column transformation rule.

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